

6. Analysis of plane truss

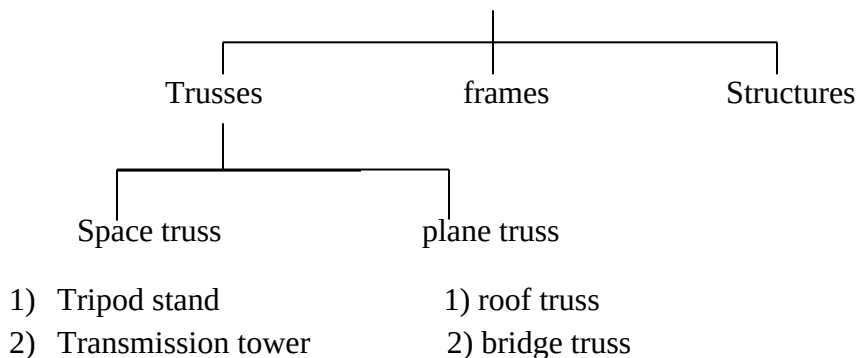
Objective of the chapter:-

The main objective of this chapter is build a strong foundation and to acquaint the students with;

- Types of trusses
- Analysis of plane truss by method of joints
- Analysis of plane truss by method of section

6.1 Introduction to Engineering structures: - These may be defined as any system of connected members built to support or transfer forces acting on them and to safely with stand these forces.

Structure



Truss: - It is a system of uniform bars or members (circular c/s, channel section etc.) joined together at their ends by riveting or welding and contracted to support loads. The members of a truss are straight members and the loads are applied only at the joints. Every members of a truss is a two force member.

Frame: - It is a structure consisting of several bars or members pinned together and in which one or more than one of its members is subjected to more than two forces. They are designed to support loads and stationary structures.

Machine: - Machines are structure designed to transmit and modify forces and contains some moving members. Here we will discuss only the plane structure that is the structures whose member lies in one plane.

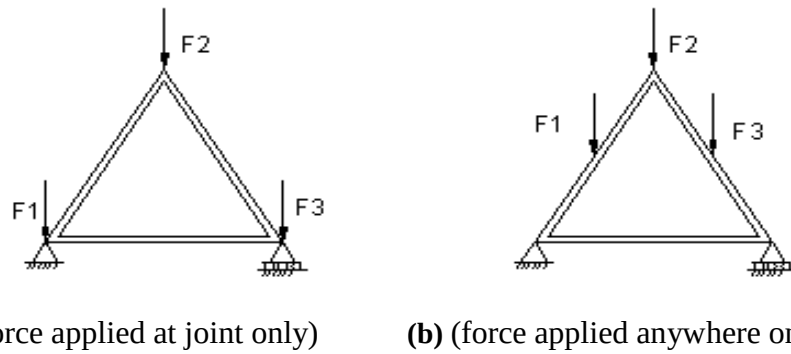
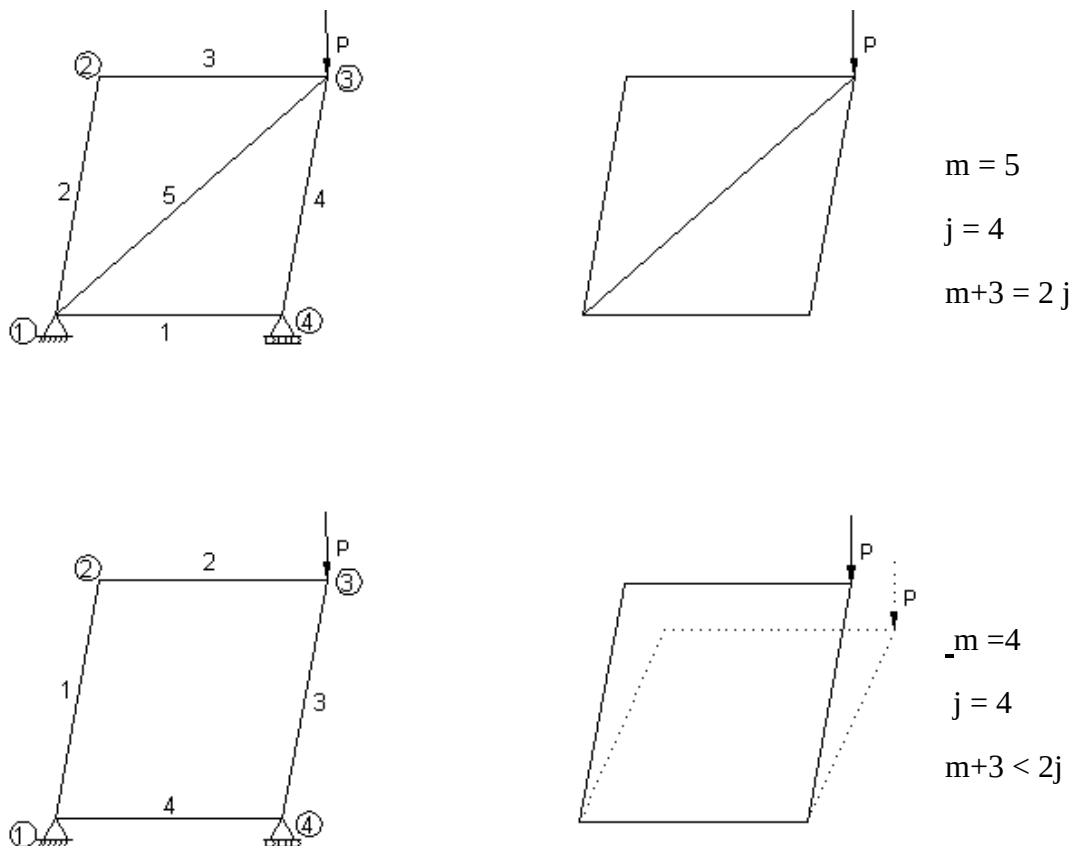


fig6.1

Rigid or Perfect truss

The term rigid, with reference to the truss, is used in the sense that, truss is the non collapsible when external supports are removed



For a truss t the rigid or perfect, the relationship between its members of joint

$m+3 = 2j$
 m = number of member in the truss.
 j = no of joints in the truss.

If $m+3 > 2j$; It means that the truss contains more members than required to be just rigid and is over rigid or statically indeterminate.

If $m+3 < 2j$; It means that the truss contains less members than required to be just rigid and is collapsible or under rigid.

Statically Determinate: A truss is statically determinate if the equations of static equilibrium alone are sufficient to determine the axial forces in the members without the need of considering their deformations.

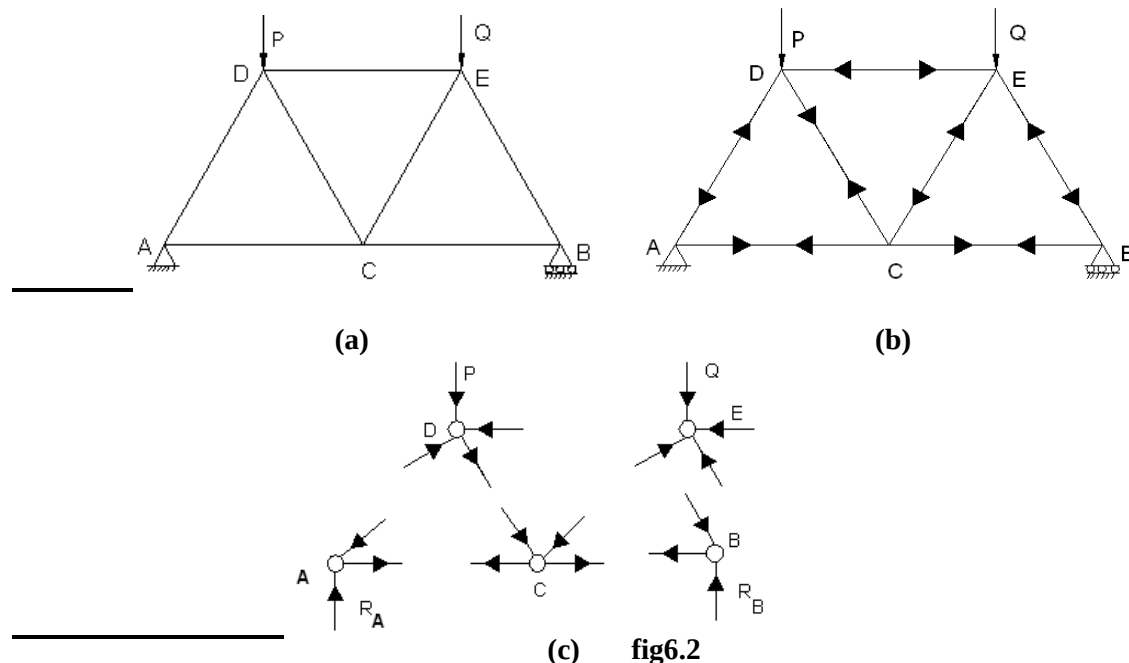
Basic Assumption for the perfect truss of a truss.

1. The joints of simple truss are assumed to be pin connection and friction less. The joints therefore cannot resist moments.
2. The loads on the truss are applied at the joint only.
3. The members of trusses are straight two force members with forces acting co-linear with the centre line of the members.
4. The weights of the members are negligible.
5. The truss is statically determinate.

6.2. Determination of axial forces in the members of a truss.

- i. Method of joints.
- ii. Method of sections.

6.2.1. Method of joints:



(c) fig6.2

(Representation of compression in the members EB. Arrows point towards the joint E & B.)

(Representation of tension in the member AC. Arrows point away from the joint A and C)
(pull at the joint)

(1) Steps for analysis:

1. Draw the free body diagram of the entire truss showing only external forces acting on it.

- Find the reactions at the supports (Roller and hinge). By considering the equilibrium of the whole truss as a rigid body, take moments of the external forces or loads applied on the truss and use the method of moment.
Note: incase of cantilever truss (fixed to the wall) where no roller support exists, it is not necessary to determine. Support reaction as above.
- Assume and mark the directions of the axial forces in the members on the diagram.
- 1st select a joint where only two unknowns are there, so that only two unknown can be determine.
- Write the equations of equilibrium by using $\sum F_x = 0$, $\sum F_y = 0$ and solve them to determine the unknown forces. On evaluation, if on axial force turns out to be positive, the assumed direction is correct, if the axial force turnout to be negative, the assumed direction is incorrect. Then the direction of axial forces should be reversed.
- Then proceed towards a joint in such a manner where two unknowns are there.
- Finally write down all the axial forces as determined above in a tabular form with their magnitude as well as nature (Tension & compression).

(2) Special Condition:

- When two members meeting at a joint are not collinear and there is no external force acting at the joint, then the forces in both the members is zero, as shown in fig. 6.3.

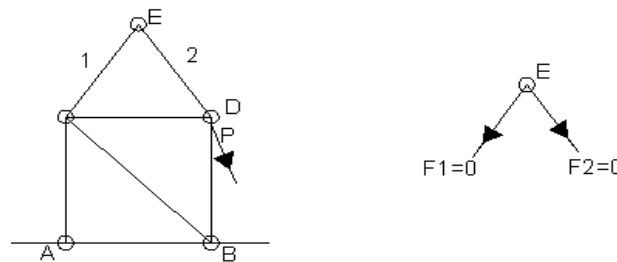


fig6.3

- When there are three members meeting at a joint, of which two are collinear and the third be at an angle and if there is no load at the joint the force in the 3rd member is zero, as shown in fig. 6.4.

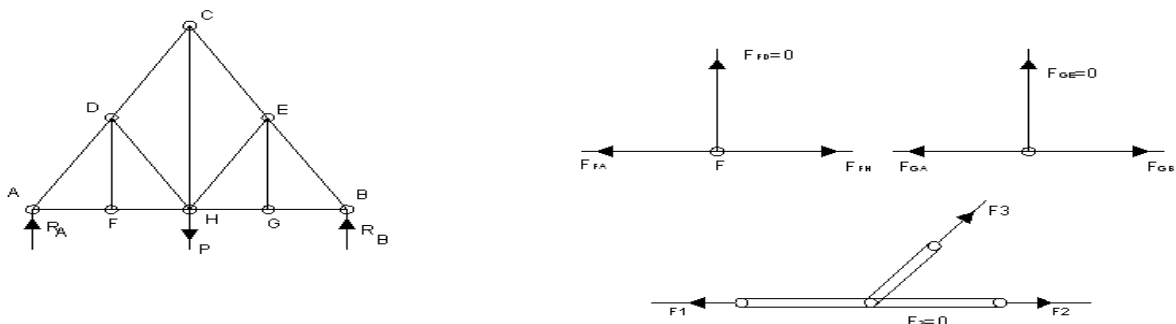


fig6.4

(3) Zero force members:

In a truss, there may be one or more members which do not carry any load. These members are called zero force members. These are used to increase the stability of the truss during construction and to provide support if the applied loading is changed. The above special conditions are the examples of zero force members.

Q: Determine the forces in all the members of the truss shown in fig. 6.5 and indicate the magnitude and nature of forces on the diagram of the truss. All inclined members are at 60° to horizontal and length of each member is 2 m.

Ans: We cannot find a joint with only two unknown forces without finding reactions.

Consider the equilibrium of the entire truss.

$$\sum M_A = 0$$

$$\Rightarrow R_D \times 4 - 40 \times 1 - 60 \times 2 - 50 \times 3 = 0$$

$$\Rightarrow R_D = 77.5 \text{ KN.}$$

$$\sum F_x = 0$$

Relation at A is vertical only.

$$\sum F_y = 0$$

$$R_A + 77.5 = 40 + 50 + 60$$

$$\Rightarrow R_A = 72.5 \text{ KN.}$$

Joint A

$$\sum F_y = 0$$

$$F_{AB} \sin 60^\circ = R_A = 72.5$$

$$F_{AB} = 83.7158 \text{ KN (Comp)}$$

$$\sum F_x = 0$$

$$F_{AB} \cos 60^\circ = F_{AE}$$

$$\Rightarrow F_{AE} = 83.7158 \cos 60^\circ = 41.8579 \text{ KN ()}$$

Joint D

$$\sum F_y = 0$$

$$\Rightarrow F_{DC} \sin 60^\circ = R_D = 77.5$$

$$\Rightarrow F_{DC} = 89.4893$$

$$\sum F_x = 0$$

$$F_{DE} - 89.4893 \cos 60^\circ = 0$$

$$\Rightarrow F_{DE} = 44.7446 \text{ KN}$$

Joint B

$$\sum F_y = 0$$

$$F_{BE} \sin 60^\circ + 40 = 0$$

$$F_{BE} = \frac{72.5 - 40}{\sin 60^\circ}$$

$$= 37.5278 \text{ (Tension)}$$

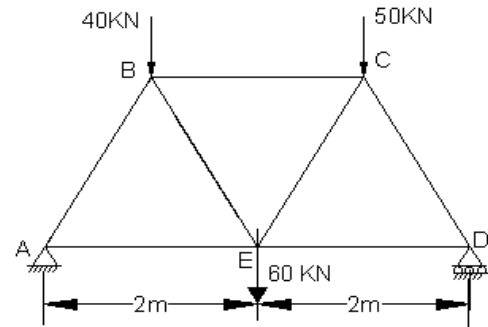
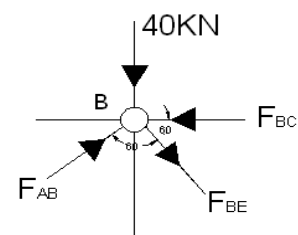
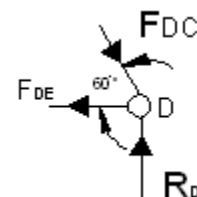
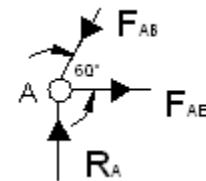
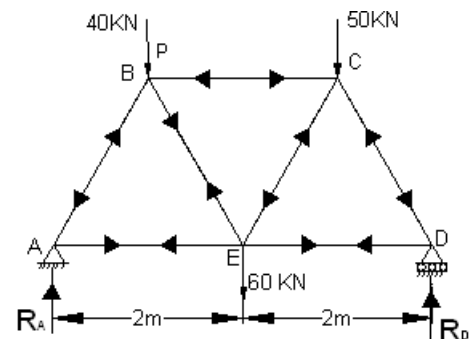


fig. 6.5

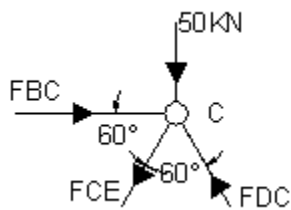


$$\sum F_x = 0$$

$$F_{BC}-F_{AB} \cos 60^{\circ}-F_{BE} \cos 60^{\circ}=0$$

$$F_{BC}=(83.7158+37.5274) \times 0.5$$

$$F_{BC}=60.6218 \text { (comp.)}$$



Joint C

$$\sum F_y = 0$$

$$F_{CE} \sin 60^{\circ}+50-F_{DC} \sin 60^{\circ}=0$$

$$F_{CE}=\frac{77.5-50}{\sin 60^{\circ}}=31.75 \text { (Tension)}$$

Now all the forces in all members are known.

Force Analysis Table

Member	Magnitude(kN)	Nature
AB	83.71	Compressive
AE	41.85	Tensile
CD	89.48	Compressive
ED	44.74	Tensile
BE	37.52	Tensile
BC	60.62	Compressive
	31.75	Tensile

Q: Determine the axial forces in all the bars of the plane truss loaded as shown in fig.6.6.

Ans. This is a cantilever type turns whose one end is fixed to the wall at point D and E and other end A is free. It is convenient to the free end A for the analysis.

From geometry ΔABC $\tan \theta = \frac{BC}{AC} = \frac{1.5}{3} = \frac{1}{2}$

$\sin \theta = \frac{1}{\sqrt{5}}, \cos \theta = \frac{2}{\sqrt{5}}$.

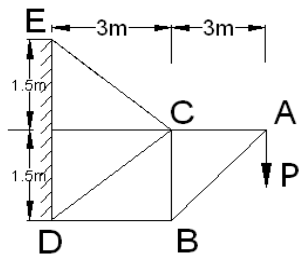


fig. 6.6

Joint A

$$\sum F_y = 0$$

$$\Rightarrow -S_2 \sin \theta - P = 0$$

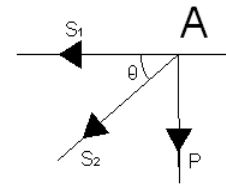
$$S_2 = \frac{P}{\sin \theta} = \frac{-P}{\frac{1}{\sqrt{5}}} = -\sqrt{5}P$$

$$\Rightarrow -2.236P = 2.236P \text{ (compression)}$$

$$\sum F_x = 0$$

$$-S_1 - S_2 \cos \theta = 0$$

$$S_1 = -(-2.236P) \frac{2}{\sqrt{5}} = 2P. \text{ (Tension)}$$



Joint B

$$\sum F_y = 0$$

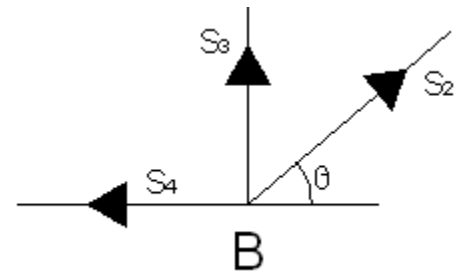
$$S_2 \sin \theta + S_3 = 0$$

$$S_3 = -S_2 \sin \theta = -(-\sqrt{5}P) \times \left(\frac{1}{\sqrt{5}}\right) = P \text{ (Tension).}$$

$$\sum F_x = 0$$

$$S_2 \cos \theta - S_4 = 0$$

$$S_4 = S_2 \cos \theta = (-\sqrt{5}P) \times \frac{2}{\sqrt{5}} = -2P = 2P \text{ (Compression)}$$



Joint C

$$\sum F_y = 0$$

$$S_6 \sin \theta - S_3 - S_5 \sin \theta = 0$$

$$\Rightarrow \frac{S_6}{\sqrt{5}} - \frac{S_5}{\sqrt{5}} = S_3$$

$$\Rightarrow \frac{S_6}{\sqrt{5}} - \frac{S_5}{\sqrt{5}} = P \text{ (1)}$$

$$\Rightarrow S_6 - S_5 = \sqrt{5}P$$

$$\sum F_x = 0$$

$$S_1 - S_6 \cos \theta - S_5 \cos \theta = 0$$

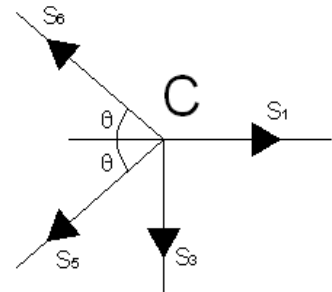
$$\Rightarrow S_5 + S_6 = \frac{S_1}{\cos \theta} = 2P \times \frac{\sqrt{5}}{2} = \sqrt{5}P \text{ (2)}$$

Subtracting equation (1) from (2)

$$S_5 + S_6 - S_6 + S_5 = \sqrt{5}P - \sqrt{5}P$$

$$\Rightarrow S_5 = 0$$

$$\text{From equation (2) } S_6 = \sqrt{5}P \text{ (Tension)}$$



Force Analysis Table

Member	Magnitude	Nature
1	2P	Tensile
2	2.236P	compressive
3	P	Tensile
4	2P	Compressive
5	0	--
6	$\sqrt{5}P$	tensile

Summary

- Truss is a system of uniform bars or members (circular c/s, channel section etc.) joined together at their ends by riveting or welding and contracted to support loads. The members of a truss are straight members and the loads are applied only at the joints. Every members of a truss is a two force member.
- In a truss, there may be one or more members which do not carry any load. These members are called zero force members.
- Determination of axial forces in the members
 - i. Method of joints.
 - ii. Method of sections

Important formulae

- The equations of equilibrium for general cases of forces acting in one plane are
 $\sum X_i = 0$ $\sum Y = 0$ $\sum (M_o)_i = 0$

6.2.2 Method of Section

In this method, the equilibrium of a portion of the truss is considered which is obtained by cutting the truss by some imaginary section.

Consider a truss as shown in fig. Cut the truss into two separate portions by passing an imaginary section through those members in which force are to be determined. The section *mn* cuts the members EF, BF and BC and the internal force in these members becomes external forces acting on the two portions of the truss.

- The equilibrium of the truss would also be in equilibrium.
- Therefore, three equations of equilibrium.

$$\sum F_x = 0, \sum F_y = 0 \text{ and } \sum M = 0$$

can be written for any one portion of the truss and can be solved to determine the three unknowns.

Following points should be noted while using the method of section.

1. The section should be passed through the members and not through the joints.
2. The section should divide the truss into two clearly separate and unconnected portions.

3. A section should cut only three members since only three unknowns can be determined from the three equations of equilibrium. However, in special cases more than three members may be cut by a section.
4. When using the moment equation, the moment can be taken about any convenient point which may or may not lie on the section under consideration.

Solved example

1. Using the method of sections, find the axial force in each of the bars 1, 2, 3 of the plane truss as shown in fig. 6.7

Ans. Since it is a cantilever plane truss, there is no need to find the support reactions at B and C.

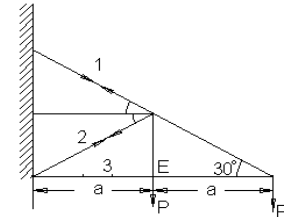
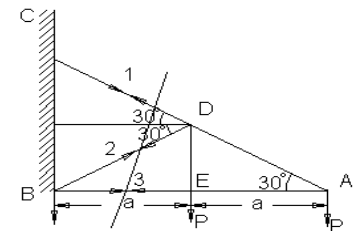


fig.6.7



Taking moment about D

$$\Sigma M_D = 0,$$

$$S_3 \times DE + P \times AE = 0$$

$$\Rightarrow S_3 = -\frac{P \times AE}{DE} = \frac{-P \times a}{a \tan 30^\circ} = -\sqrt{3}P = \sqrt{3}P \text{ (comp}^n\text{)}$$

$$\Sigma M_B = 0,$$

$$S_1 \sin 30^\circ \times BA - P \times BA - P \times BE = 0$$

$$\frac{S_1}{2} \times 2a = P \times 2a + P \times a$$

$$\Rightarrow S_1 = 3P \text{ (Tension)}$$

Taking the vertical forces on RHS

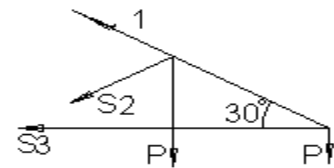
$$\Sigma F_y = 0,$$

$$S_1 \sin 30^\circ - S_2 \sin 30^\circ - P - P = 0$$

$$\Rightarrow \frac{S_1}{2} - \frac{S_2}{2} = 2P$$

$$\Rightarrow S_2 = \frac{S_1}{2} - 2P = \frac{3P}{2} - 2P = -0.5P$$

$$\Rightarrow S_2 = -P = P \text{ (Comp}^n\text{)}$$



2. Using the method of sections, find the axial forces in each of bars 1, 2, 3 of the plane truss as shown in fig 6.8.

Ans.

Section mn is taken cutting the members 1, 2 and AE.

Taking moment about E

$$\Sigma M_E = 0$$

$$S_2 \times ED + 5 \times AE = 0$$

$$\Rightarrow S_2 = \frac{-5 \times AE}{ED} = \frac{-5 \times 3}{1.5} = -10 \text{ kN} = 10 \text{ kN (Comp}^n)$$

Taking all vertical forces acting on the section

$$\Sigma F_v = 0,$$

$$S_1 - 5 = 0$$

$$\Rightarrow S_1 = 5 \text{ kN (Tension)}$$

The second section is line pq intersecting the bars 2, 3, CE

Taking moment about C,

$$\Sigma F_c = 0$$

$$S_2 \times 3 + S_3 \cos \theta \times 3 + 5 \times 6 = 0$$

$$-10 \times 3 + S_3 \cos \theta \times 3 + 30 = 0$$

$$\Rightarrow S_3 \cos \theta = 0$$

$$\Rightarrow S_3 = 0$$

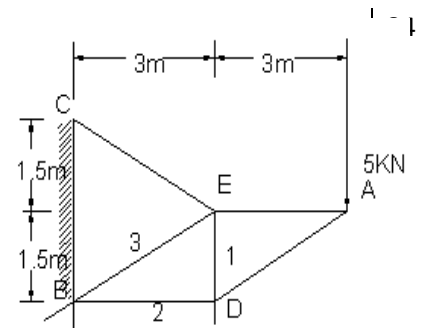
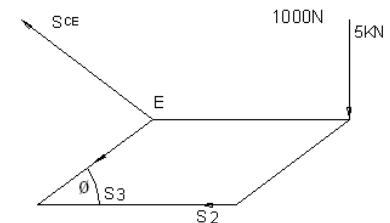
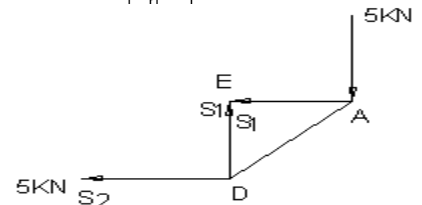
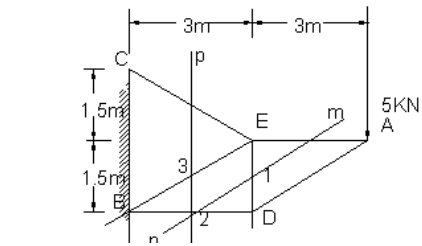


fig.6.8



3. Using the method of sections, find the axial forces in each of bars 1, 2, 3 of the plane cantilever truss as shown in fig 6.9

Ans. Considering the equilibrium of section 1, 2, 4

The R.H.S

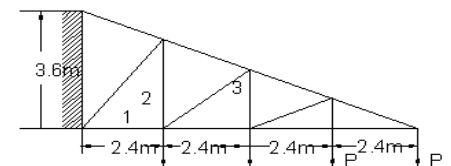


fig.6.9

$$\frac{AB}{BC} = \frac{DE}{EC}$$

$$\Rightarrow DE = \frac{3.6}{9.6} \times 7.2 = 2.7\text{m.}$$

Considering equilibrium of R.H.S

$$\Sigma M_D=0$$

$$S_1 \times 2.7 + P \times 2.4 + P \times 4.8 + P \times 7.2 = 0$$

$$\Rightarrow S_1 = -5.33P = 5.33P \quad (\text{Comp}^n)$$

Taking moment about C $\Sigma M_C = 0$

$$S_2 \times 2.7 - P \times 7.2 - P \times 4.8 - P \times 2.4 = 0$$

$S_2=2P$ (Tension)

$$\begin{aligned} FG &= \frac{AB}{AC} \times GC \\ &= \frac{3.6}{9.6} = 1.8 \end{aligned}$$

$$\Delta EFG, \quad (\tan \theta) = \frac{1.8}{2.4}$$

$$\Rightarrow \theta = 36.56^\circ$$

Taking moment about C, $\Sigma M_C=0$

$$S_3 \cos 36.86^\circ \times 1.8 + S_3 \sin 36.86^\circ \times 4.8 + P \times 4.8 +$$

P ~~x~~2.4= 0

$$\Rightarrow 4.32 S_3 = -7.2P$$

$$\Rightarrow S_3 = -1.67P = 1.67 \text{ (comp}^n\text{)}$$

